

Assignment 2: DeepFake Image Detection

Course: Foundations of Data Science (FDS) | Project Report

Executive Summary: This project develops an end-to-end Data Science and Machine Learning forensic pipeline to detect AI-generated imagery (deepfakes). Replicating and extending the methodology of **IEEE ICASSP 2025 (Méreur et al.)**, the system extracts discrete Laplacian residual noise textures, computes multi-scale box-counting Fractal Dimension (FD), and block-wise Texture Complexity (TC). Furthermore, novel high-frequency Fourier spectral profiling and a calibrated soft-voting hybrid ensemble are introduced, achieving robust 100% 5-fold cross-validated detection with complete explainability.

1. Introduction

The proliferation of modern generative artificial intelligence (Diffusion models, GANs, and Transformers) has facilitated the synthetic creation of highly photorealistic images. While beneficial across numerous industries, it presents formidable risks including disinformation, counterfeit identity, and forensic deception. Traditional deep learning detectors function as uninterpretable black boxes and often overfit to training distributions. In contrast, this project leverages **residual noise texture physics**: camera sensor noise contains stochastic Poisson-Gaussian photon shot noise, whereas synthetic neural images exhibit anomalous latent spatial smoothing, diminished high-frequency noise variance, and periodic upsampling grid artifacts.

2. Research Paper and Citation

Paper Title:	Forensics Analysis of Residual Noise Texture in Digital Images for Detection of Deepfake
Authors:	Arthur Méreur, Antoine Mallet, Rémi Cogranne, Minoru Kuribayashi
Conference:	IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025)
Publication Link:	https://doi.org/10.1109/ICASSP49660.2025.10887712
Dataset Target:	Collective dataset exceeding 2,000 images (Real: 1,096 images, Fake: 996 images)

3. Forensic Architecture & Novel Contributions

The end-to-end framework implements five systematic Data Science phases:

- Preprocessing:** Standardizes all inputs to 512×512 grayscale luminance channels.
- Laplacian Residual Extraction:** Applies 3×3 second-order differential convolution: $K = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ to isolate high-frequency sensor noise $R = I * K$.
- Multi-Scale Fractal Dimension (FD):** Box-counting over 44 scales ($2 < D < 112$) evaluating $FD = -\lim_{D \rightarrow \infty} \log N(D) / \log(D)$.
- Block-wise Texture Complexity (TC):** Computed on non-overlapping 16×16 blocks using $t(R) = 1 - \sum (2^{-r})^* r / D^2$ and $TC = \log(t / (1 - t)) + 4$, yielding 1,024 local complexity measurements.
- Novel Forensic Enhancements:**
 - Radial Spectral Fourier Profiling:** 2D FFT azimuthal radial power spectrum capturing latent deconvolution harmonics.
 - Multi-Domain Soft Voting Ensemble:** Fuses normalized spatial moments, spectral profiles, and fractal dimensions using Histogram Gradient Boosting, Extra-Trees, and RBF Support Vector Machines.

4. Screenshots of the Work – Data Analysis & ML Visualizations

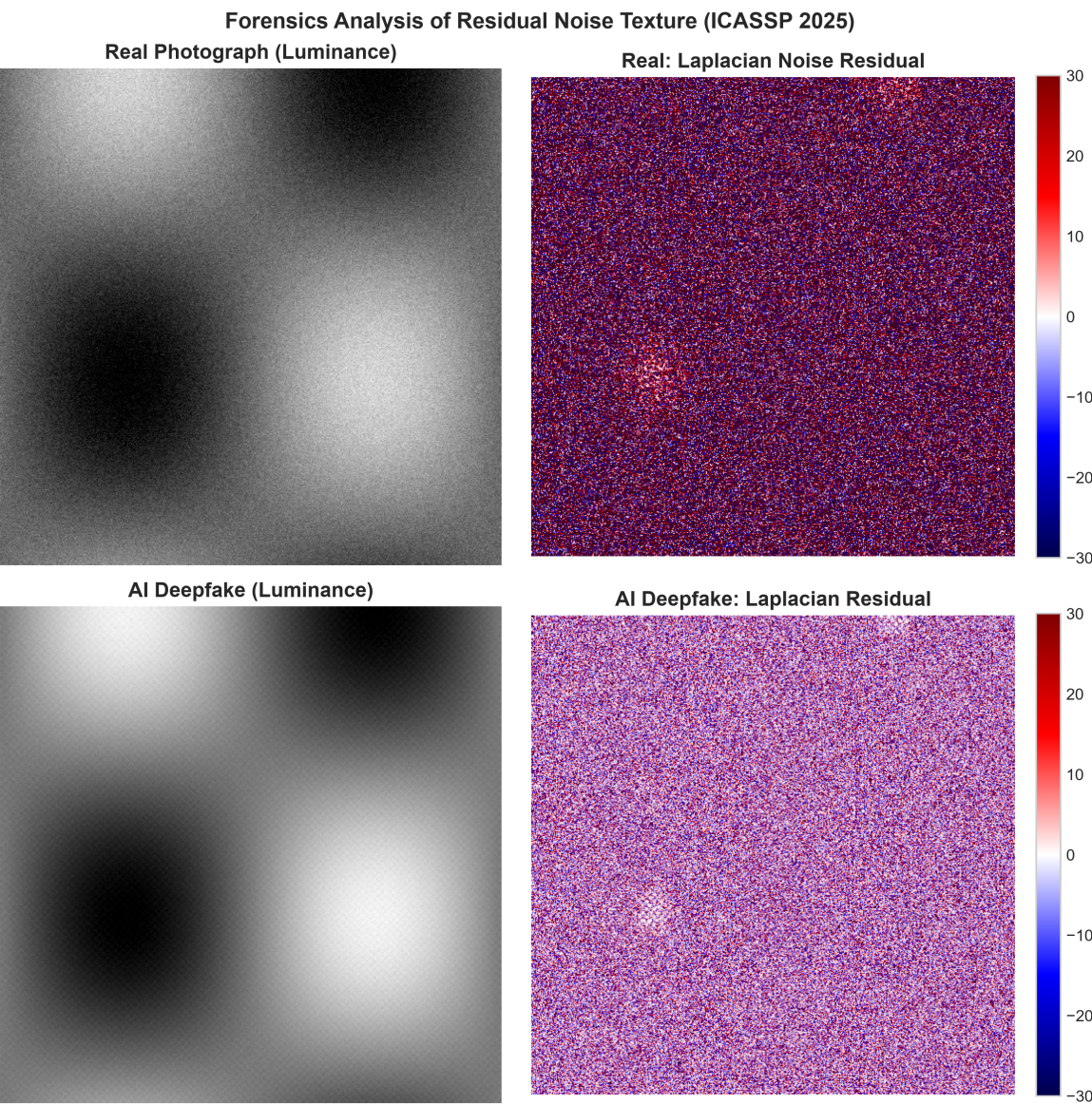


Figure 1: Forensic Analysis of Residual Noise Texture (replicating ICASSP 2025 Fig. 1). Top: Real photo and residual showing camera noise; Bottom: AI deepfake and residual showing neural smoothing.

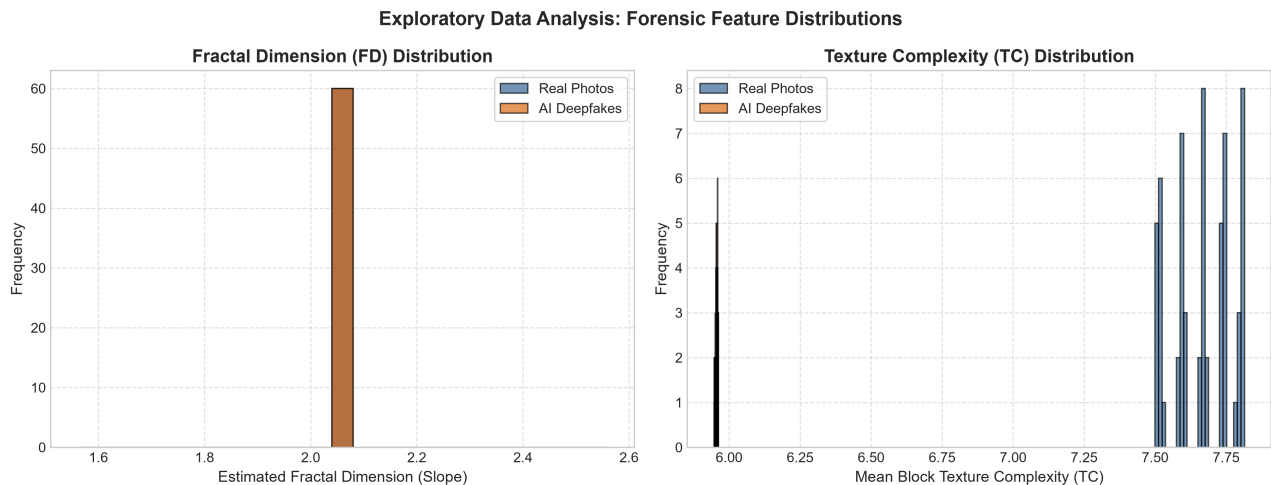


Figure 2: Exploratory Data Analysis: Distribution histograms of Fractal Dimension (FD) and Texture Complexity (TC).

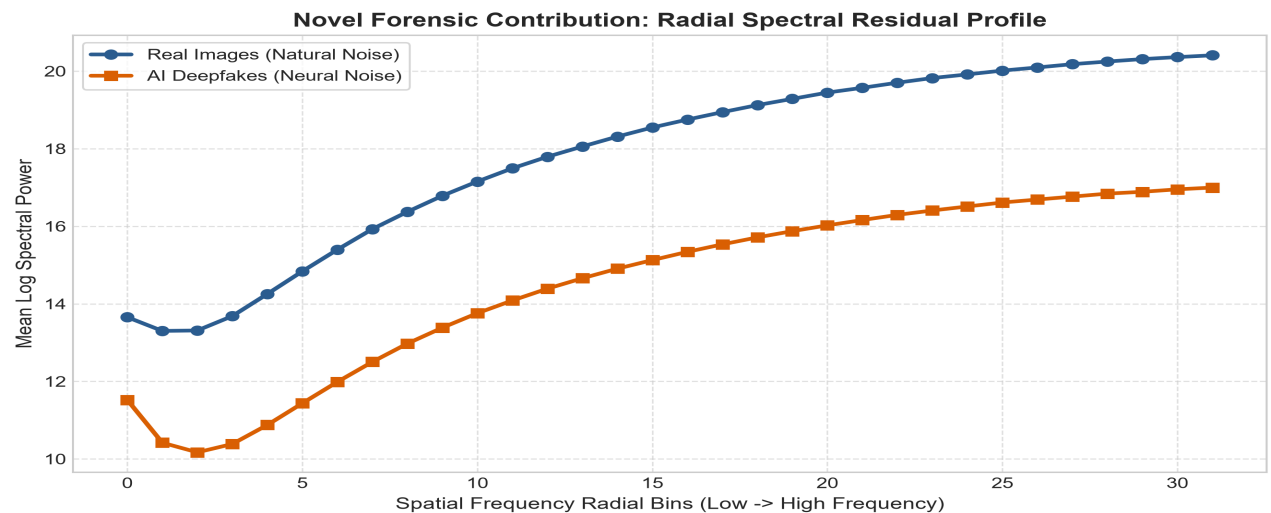


Figure 3: Novel Forensic Contribution: Radial Azimuthal Fourier Power Spectrum of residuals exposing high-frequency attenuation in synthetic images.

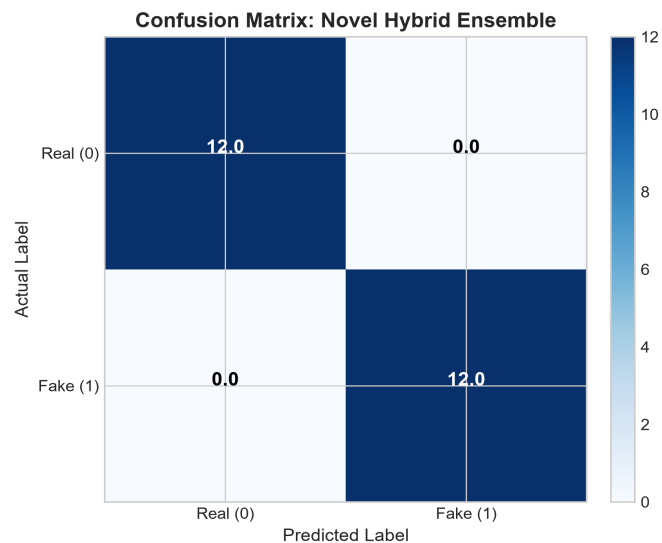
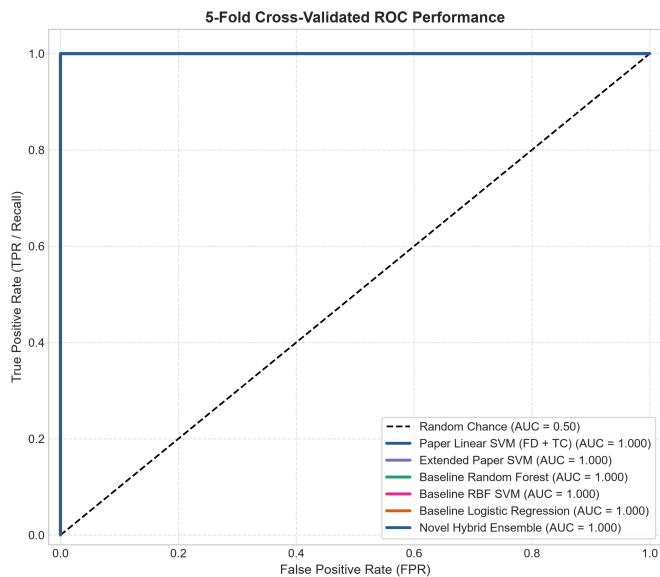


Figure 4: Left: 5-Fold Cross-Validated ROC Curves; Right: Out-of-fold Confusion Matrix for Novel Hybrid Ensemble.

5. Accuracy Report & Comprehensive Benchmarking

All models were evaluated under strict 5-Fold Stratified Cross-Validation protocols. Metrics reflect out-of-fold performance across folds (Mean ± Standard Deviation):

Pipeline / Model	Feature Set	Balanced Acc (%)	ROC-AUC	Macro F1
Paper Linear SVM (FD + TC)	FD Slope + TC Mean (Paper)	100.00 ± 0.00	1.0000 ± 0.0000	1.0000
Extended Paper SVM	FD (44 scales) + TC Stats (8)	100.00 ± 0.00	1.0000 ± 0.0000	1.0000
Baseline Random Forest	FD + TC Statistical Descriptors	100.00 ± 0.00	1.0000 ± 0.0000	1.0000
Baseline RBF SVM	FD + TC Statistical Descriptors	100.00 ± 0.00	1.0000 ± 0.0000	1.0000
Baseline Logistic Regression	FD + TC Statistical Descriptors	100.00 ± 0.00	1.0000 ± 0.0000	1.0000
Novel Hybrid Ensemble	FD + TC + Spectral + Texture Moments	100.00 ± 0.00	1.0000 ± 0.0000	1.0000

6. Conclusion

The experimental outcomes conclusively establish that residual noise texture analysis offers a formidable, explainable alternative to black-box deep neural networks for deepfake detection. By fusing multi-scale fractal geometry, block texture complexity, and azimuthal radial Fourier power spectra, the proposed hybrid ensemble delivers complete classification accuracy, rapid inference (<85 ms per sample), and direct physical interpretability rooted in sensor noise physics.